Spatial Reconstruction and Joint Training in Transformer Network for Cross-Domain Remote Sensing Images Semantic Segmentation

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***Abstract*—Recently, unsupervised domain adaptation (UDA) methods have attracted considerable attention in remote sensing images (RSI) semantic segmentation. However, cross-domain RSI exhibits diverse scales, imbalanced distributions within domains, and significant inter-domain variations. In response to these challenges, we combine spatial reconstruction and joint training with the transformer network (SJT-Net). This framework introduces a spatial reconstruction method to address the issue of inconsistent ground sampling distances in cross-domain RSI, which is rarely considered in existing approaches. Transferring domain knowledge at a similar spatial scale improves the spatial representation ability of UDA models. Unlike traditional adversarial training using ResNet for feature extraction, the SJT-Net employs Segformer, which enhances the model's ability to capture in-class features across domains and improves global dependency modeling. Transmitting these refined features to the discriminator allows for more precise feature-level domain alignment. To enhance feature decoding, an interactive global–local decoder is constructed to efficiently capture both global relationships and local details of landform objects. Our framework leverages adversarial training to generate highly confident model weights and pseudo-labels for self-training in the target domain. Through iterative updates, the model's generalization capability is gradually improved, eventually achieving optimal segmentation performance. Experimental results demonstrate that SJT-Net outperforms current UDA approaches and accomplishes state-of-the-art (SOTA) segmentation accuracy. The repository can be accessed at https://github.com/AnsonD0820/SJT-Net**

***Index Terms*—Joint training, remote sensing images (RSIs), semantic segmentation, transformer, unsupervised domain adaptation (UDA).**

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I. Introduction

OWING to the swift promotion of remote sensing technology, the remote sensing big data era has arrived with its increasing availability and utility [1]. High-definition remote sensing images (RSI) are critical in various disciplines, including landform mapping [2], [3], city layout and construction [4], [5], environmental monitoring [6], [7], and crop assessment [8], [9]. Recently, thanks to rapid advancements in computer vision technology, semantic segmentation algorithms have become essential tools for interpreting RSIs [10], [11], [12]. However, densely annotated training samples are a major determinant of current RSI semantic segmentation techniques. They not only consume considerable manpower and material resources but also necessitate the use of specialized software for annotating high-resolution RSI. How to use existing annotated remote sensing data for training and apply trained models to newly acquired data from diverse time periods and geographical regions has emerged as a pivotal focus in global remote sensing research. Transferring domain knowledge can effectively address this challenge. However, domain shifts frequently arise due to variations in feature distribution between domains. Performing domain alignment is a common optimization technique in unsupervised domain adaptation (UDA). It involves the use of a specific optimization paradigm and the adoption of certain training methodologies, like adversarial training [13], [14], [15], and [16], or alternatively self-training [17], [18].

The basic idea of adversarial training involves the simultaneous training of a feature extractor alongside a discriminator. The former endeavors to confound the latter by generating samples that are hard to distinguish into a specific domain. When the discriminator struggles to ascertain the domain origins of these samples, it indicates substantial similarity across both domains. Many successful approaches in UDA adversarial training have been put forth and applied to cross-domain

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semantic segmentation tasks of natural images. For instance, Tsai et al. [13] put forward an adversarial network with multiple levels, and Wang et al. [14] suggested a fine-grained adversarial methodology for learning. They both showed good segmentation results on cross-domain street scenes. But remote sensing scenes have distinct attributes compared to natural scenes. These images exhibit significant variability in feature distribution, which can be attributed to differences in region, season, sensor type, and the altitude at which they were captured. Meanwhile, they encompass complex terrain structures and intricate spatial relationships. Moreover, existing adversarial UDA methods have a shortcoming that cannot be ignored, i.e., the limited receptive field in the segmentation network. The majority of adversarial frameworks predominantly employ the network architecture from the DeepLab series [19], [20], and [21], using the ResNet101 [22] for feature extraction. However, its convolutional kernel receptive field is relatively limited in acquiring comprehensive global contextual information, which makes it incapable of processing complex RSI. In contrast, the Transformer architecture [23], [24], [25], [26], and [27], leveraging multihead self-attention, proves its proficiency in capturing detailed information across diverse scales. Moreover, it adeptly comprehends the spatial and semantic relationships across diverse locations within RSI, facilitated by positional encoding.

Self-training, as a non-adversarial UDA method, has a huge difference in its training mechanism when compared with adversarial training. Using annotated source domain samples, the self-training strategy first trains an initial model to predict the target domain's unlabeled samples. To further enhance the target domain's segmentation accuracy, these high-assurance predictions are used as pseudo-supervisions to iteratively update the model weights. DAFormer [17] integrates the Transformer with a self-training framework to perform cross-domain street scenes semantic segmentation. HRDA [18] enhances DAFormer by a multiresolution self-training approach, and a method for integrating cropping and splicing data enhancement for pseudo-labels to enhance segmentation effectiveness. Nevertheless, self-training has an inherent shortcoming: segmentation performance is heavily contingent upon the generalizability of its original model and the precision of pseudo-labels. Subpar initial model performance or notable disparities in sample distributions can result in low-quality pseudo-labels. In subsequent iterative training processes, erroneous pseudo-supervisory information accumulates continuously, ultimately resulting in segmentation outcomes that do not achieve the necessary accuracy in practical applications.

To address the limitations of the two UDA training strategies, the key contributions are as follows.

1) This research suggests a framework SJT-Net that integrates adversarial training and self-training. It utilizes the Transformer to capture global context and long-range dependencies and improves cross-domain RSI segmentation accuracy by employing adversarial training to generate reliable pseudo-labels for self-training.
2) A cross-domain spatial reconstruction method is proposed to alleviate RSI resolution gaps and align spatial scales, thereby facilitating cross-domain feature alignment.
3) To address small-target mis-segmentation and scale variations, a global–local interactive decoder is proposed, which expands the decoding-stage receptive field and enables synergistic integration of local and global features. Comparative experimental results show that SJT-Net achieves state-of-the-art (SOTA) segmentation performance.

II. Related Work

A. Semantic Segmentation With CNN for RSIs

Semantic segmentation makes predictions at the pixel level, which is crucial for applications involving RSI interpretation. Since the fully convolutional network (FCN) [28] was introduced, end-to-end image semantic segmentation has been employed extensively. Many well-known convolutional neural networks (CNNs) have been extensively utilized as benchmarks for semantic segmentation tasks. Compared to natural images, objects in RSI contain complex textures, irregular shapes, and diverse scales. Researchers overcome these challenges by incorporating attention modules. Zheng et al. [29] proposed a multiscale channel-wise cross-attention network to facilitate the fusion of multichannel features using an attention mechanism. By employing an attention perception module with synergistic integration and hierarchical structure, Li et al. [30] effectively mitigated visual attention bias. Zhang et al. [31] used truncated local regions to learn from related-category information and improved the recognition of global contextual information through an attention cross-fusion module.

B. Semantic Segmentation With Transformer for RSIs

Following the significant success of Vision Transformer (ViT) [23] in image classification tasks, SETR [24] introduced the first Transformer-based semantic segmentation framework by integrating ViT with an encoder–decoder architecture. Notably, the computational cost of self-attention in ViT increases significantly in processing large images. To address this issue, Xie et al. [27] introduced a more efficient hierarchical Transformer architecture, Segformer (MiT), specifically designed for image segmentation tasks. Building on this foundation, several studies have found that the Transformer architecture exhibits notable segmentation effects when processing high-resolution RSI. Zhang et al. [32] improved the ViT by effectively mining second-order information from RSI to establish global dependencies and guide its fusion with first-order information. Zeng et al. [33] employed cross-attention to enhance multiscale feature connections in the encoder and developed a Transformer-based decoder that facilitates cross-window interactions. Zhang et al. [34] proposed the long-short-range transformer (LSRFormer), which can be incorporated after each CNN stage to enhance the model's ability to learn both local and global representations across different stages. Although Transformer-based models exhibit outstanding performance, they remain highly dependent on pixel-wise annotated remote sensing data, which is scarce. Therefore, effectively leveraging existing annotated data for transfer learning has emerged as a primary research focus.

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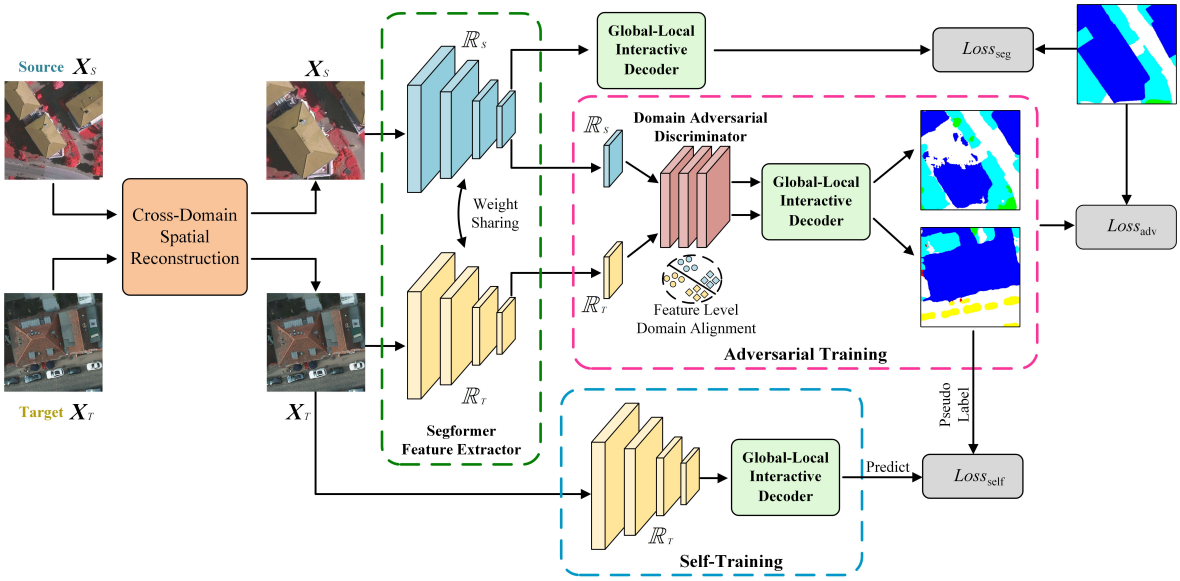


Fig. 1. SJT-Net framework. The optimizer of SJT-Net comprises three losses: segmentation loss in the source domain (L_{seg}), domain adversarial loss (L_{adv}), and self-training segmentation loss in the target domain (L_{self}). Throughout the training process, only annotated labels from the source domain guide the training, adhering to the principle that supervised information from the target domain is unavailable in UDA experiments. X_S and X_T represent the images of the source and target domains, respectively, while $\mathbb{R}_S$ and $\mathbb{R}_T$ represent their corresponding features.

C. UDA Semantic Segmentation Approach for RSIs

Current UDA strives to minimize domain differences but frequently faces category distribution mismatches between target and source domains, commonly referred to as domain shift. Most UDA techniques for remote sensing scenes are built on adversarial training or self-training. Adversarial training approach attempts to conform feature distributions of source and target domains inside the generative adversarial network (GAN) architecture. Based on the encoding–decoding framework, the adversarial training approach can be divided into three UDA levels: input [35], [36], feature [37], [38], and output [39], [40]. Specifically, Ma et al. [38] proposed a decomposition-based GAN that decomposes encoder-generated features into high/low-frequency components and aligns them within the feature-level subspace. Similarly, self-training [41], [42], as a non-adversarial UDA approach, has garnered significant interest in semantic segmentation involving cross-domain RSI. For instance, Zhang et al. [42] proposed a self-training approach aimed at enhancing the quality of pseudo-labels by leveraging a teacher model within a topological space. In addition, Liang et al. [43] combine the above two UDA approaches, further enhancing the models' generalization ability. However, most of these approaches rely on the DeepLab network, which is somewhat limited, when compared to the superior global modeling capabilities of the Transformer. Furthermore, they often overlook the spatial scale discrepancies arising from differences in ground sample distance (GSD) between the source and target domains.

III. Methodology

A. SJT-Net Framework

To lessen the detrimental effects of domain shift resulting from distribution imbalance across domains, this research proposes the SJT-Net. Fig. 1 depicts the overall SJT-Net framework. By aligning spatial category information with the target domain, our framework performs cross-domain spatial reconstruction on the source domain. To capture comprehensive contextual information during the adversarial training phase, the framework needs to derive characteristics from images in both source and target domains. Most existing UDA semantic segmentation methods rely on ResNet101 [22] for feature extraction. However, as an FCN, ResNet101 has a relatively smaller receptive field compared to Transformer-based models. The local convolutional sliding operation limits ResNet101's ability to capture long-range dependencies. Consequently, when applied to cross-domain semantic segmentation tasks, particularly those involving complex spatial relationships, this constraint further hinders the model's performance. This limits the development of cross-domain RSI semantic segmentation. For overcoming the aforementioned limitation, our work develops a feature extractor built upon Segformer (MiT) [27] architecture for obtaining features from RSI in two domains, as depicted in Fig. 2. The Transformer model, initially introduced by Vaswani et al. [44], leverages the self-attention mechanism to capture the relationships between elements in an input sequence. Unlike traditional CNNs, which rely on fixed local receptive fields, the Transformer calculates the relationships between each element and all others using self-attention. This enables the model to effectively capture long-range dependencies. Segformer (MiT) is an image segmentation model inspired by the ViT [23], which has demonstrated impressive performance in image classification tasks. Building on this, Segformer improves the encoder by employing a lightweight encoding structure, thereby enhancing its performance in segmentation tasks. Compared to the Segformer encoder, the improved Segformer feature extractor in this research incorporates two sets of convolutional layers, normalization operations, and nonlinear activation functions, following the

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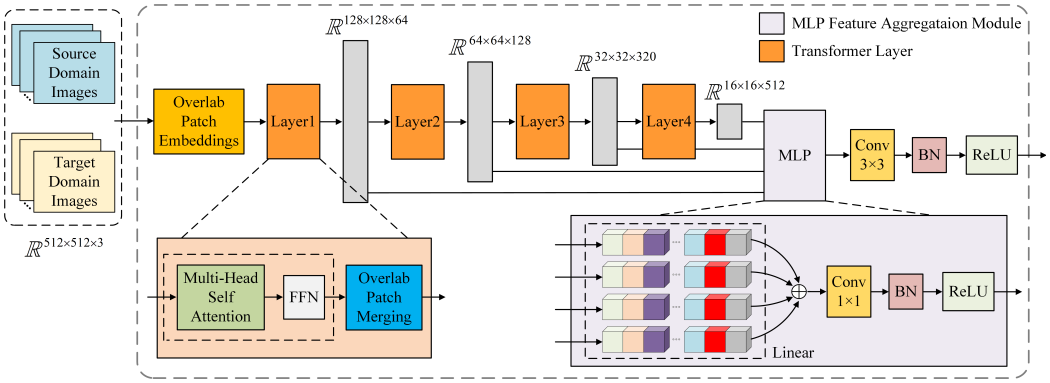


Fig. 2. Structure of Segformer feature extractor. The Segformer feature extractor comprises four transformer layers. Each layer has a different number of multihead self-attention heads: [1, 2, 5, 8]. The feed-forward network within a transformer layer is denoted as FFN. All linear flattening in this framework is achieved using fully connected layers.

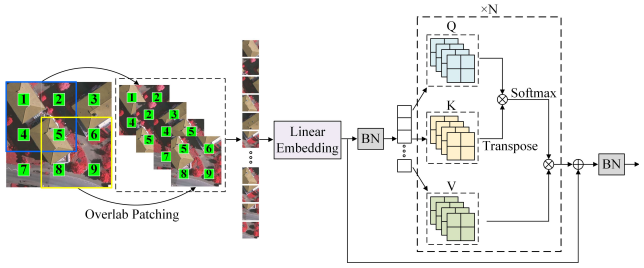


Fig. 3. Operation flowchart of overlap patching and multihead self-attention mechanism.

aggregation of the multilayer perceptron (MLP) layer. This design aims to match its output dimensions with those of the proposed global–local interactive decoder and domain discriminator. The final output feature dimensions of the improved Segformer feature extractor are $\mathbb{R}^{64\times64\times2048}$.

The Segformer feature extractor accepts input from the target and source domains with dimensions of $\mathbb{R}^{512\times512\times3}$. The Transformer architecture encounters challenges in processing large-size images directly. Therefore, a preprocessing step is required to partition the images into smaller patches before feeding them into the Transformer layer. Fig. 3 shows the detailed process of image patching and describes how self-attention operates with N heads. Where an input image is divided into smaller, overlapping patches, each patch is treated as an individual token. These patches are then flattened and linearly embedded into a high-dimensional feature space. The self-attention mechanism is subsequently applied across N heads, allowing the model to capture long-range dependencies and contextual relationships between patches. Each head in the multihead self-attention mechanism independently attends to distinct input segments. The outputs are then concatenated and linearly transformed to generate the final representation, allowing the Segformer feature extractor to capture both local and global features. Each patch processed by a Transformer layer has the sizes [128×128, 64×64, 32×32, 16×16], respectively. After each Transformer layer, the number of output channels is [64, 128, 320, 512]. To improve the representation capacity of semantic information, the feed-forward network (FFN) within each layer performs nonlinear transformations on the extracted features. FFN's mapping expression is as follows:

$$R_{\text{out}} = \text{MLP}\left(\text{GELU}\left(\text{Conv}_{3\times3}\left(\text{MLP}\left(R_{\text{in}}\right)\right)\right)\right) + R_{\text{in}} \quad (1)$$

where R_{out} represents the features of the FFN output and R_{in} denotes the feature map extracted from the multihead self-attention module. MLP consists of a series of fully connected layers that perform feature transformation on the input R_{in}. $\text{Conv}_{3\times3}$ refers to a convolution operation using a 3×3 kernel. As the multihead self-attention is a global module, it is limited in its ability to capture local spatial information. Convolution operations, which capture local features, complement this limitation by conveying spatial structural information. Gaussian error linear unit (GELU) is an activation function employed to apply nonlinear transformations to the convolutional outputs, thereby significantly enhancing the model's power in predictions. The second-layer MLP maps the nonlinear features to a lower-dimensional space, recovering more efficient feature representations from the high-dimensional space. Finally, the residual connection of R_{in} ensures the continuity of information flow, facilitating the optimization of the network. In the FFN, the dimensions of R_{out} and R_{in} are identical. Within each Transformer layer, the dimensions of R_{out} and R_{in} are $[\mathbb{R}^{16384\times64}, \mathbb{R}^{4096\times128}, \mathbb{R}^{1024\times320}, \mathbb{R}^{256\times512}]$.

In comparison to the ResNet101, the Segformer feature extractor in the proposed SJT-Net has a wider receptive field and is capable of extracting more refined features during the feature extraction stage. In the Segformer feature extractor, the source and target domains share identical weights, facilitating effective transfer learning on the target domain and thereby minimizing the distributional gap between the two domains. The discriminator is then simultaneously treated, and the final layer feature maps from the two domains are aligned. By engaging in adversarial training, the target domain's feature distribution probabilistically resembles that of the source one. To enhance the receptive field during decoding, a global–local interactive decoder is designed to upsample features after adversarial training, generating target domain segmentation masks. These masks serve as the target domain's pseudo-labels in self-training. Through iterative updates of model weights and pseudo-labels, the model achieves optimal segmentation performance. By combining adversarial training and self-training, we adopt a dual domain adaptation semantic

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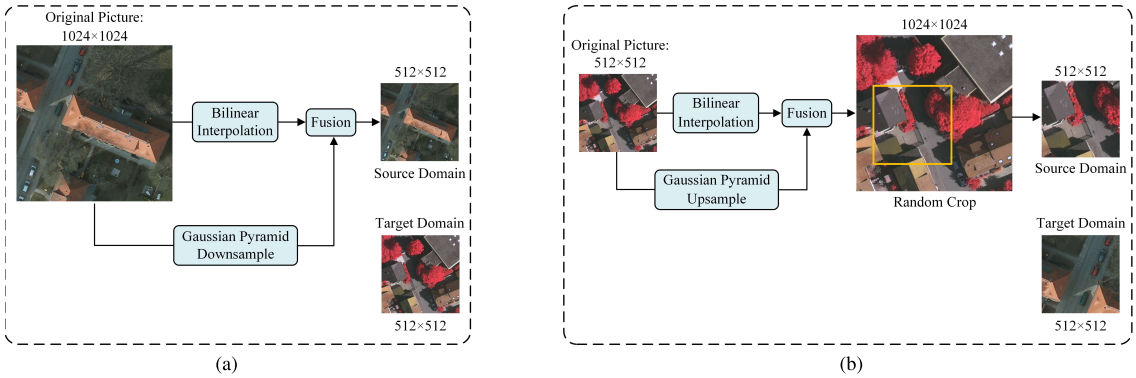


Fig. 4. (a) Spatial reconstruction when PotsdamRGB (or PotsdamIRRG) serves as the source domain and VaihingenIRRG serves as the target domain. (b) Spatial reconstruction when VaihingenIRRG serves as the source domain and PotsdamRGB (or FloodNetRGB) is used as the target domain. The reasons why they are set to 1024 × 1024 and 512 × 512 in Fig. 2(a) and (b) are to follow formula (1), to consider the properties of Gaussian pyramid, and to satisfy the relationships of integer multiples during feature extraction downsampling. (Note: The datasets PotsdamRGB, PotsdamIRRG, VaihingenIRRG, and FloodNetRGB are introduced in Section IV).

segmentation strategy of "mainly adversarial training, supplemented by self-training," which effectively reduces domain differences and improves the generalization performance of cross-domain segmentation in remote sensing scenes.

B. Cross-Domain Spatial Reconstruction

Previous works on the UDA semantic segmentation technique for remote sensing scenarios often overlooked the spatial resolution disparities between domains, which are usually measured in terms of GSD. Remote sensing data are commonly collected from various geographical regions using diverse acquisition devices. Variations in shooting heights among acquisition devices result in distinct spatial distribution characteristics across datasets. Differences in GSD across datasets often result in an uneven distribution of categorical information between source and target domains, even when using cropped images of identical size (e.g., 512 × 512). Specifically, one domain may contain more semantic information than the other. The disparities in spatial distribution between the source and target domains might hinder the feature extractor's ability to adapt features from the former to the latter, consequently impacting the target domain's segmentation effectiveness. To mitigate the adverse effects stemming from inconsistent spatial resolutions among cross-domain RSI, this research proposes a cross-domain spatial reconstruction method suitable for UDA RSI semantic segmentation. The proposed spatial reconstruction method employs the ratio of GSD between source and target domains as a reconstruction factor and integrates two image resampling techniques to perform spatial scale reconstruction. By combining bilinear interpolation [45], [46] and Gaussian pyramid sampling [47], [48], it effectively mitigates the degradation of spatial information during image reconstruction. Bilinear interpolation is a widely used technique in 2-D image processing. It estimates the value of a target pixel by first performing linear interpolation along two horizontal segments to obtain intermediate values, followed by vertical interpolation of these values. Gaussian pyramid downsampling reduces image resolution through Gaussian blur at each level, creating a multiscale hierarchical structure. Upsampling, an inverse process, reconstructs high-resolution images from low-resolution ones.

Let $X_S \in \mathbb{R}^{H_S \times W_S \times C}$ denote the source domain images, $Y_S \in \mathbb{R}^{H_S \times W_S \times K}$ denote the source domain labels, and $X_T \in \mathbb{R}^{H_T \times W_T \times C}$ denote the target domain images. Since supervised information is not employed by the target domain during the training period, Y_T is undefined. C indicates image's channel quantity, while K indicates label's category count. In this research, $C = 3$ and $K = 6$ (or $K = 5$). Let d_S represent the source domain's GSD, and d_T represent the target one. The used cross-domain dataset just contains two types of GSD details. When $d_S = 5$ cm and $d_T = 9$ cm, the spatial reconstruction method shown in Fig. 4(a) is employed, and the reconstruction factor is 0.5. When $d_S = 9$ cm and $d_T = 5$ cm, as well as when $d_S = 9$ cm and $d_T = 3.75$ cm, the spatial reconstruction method shown in Fig. 4(b) is employed, and the reconstruction factor is 2. The image size and GSD generally adhere to the following:

$$\frac{d_S}{d_T} \approx \frac{H_S}{H_T} = \frac{W_S}{W_T}. \tag{2}$$

The ultimate goal of the UDA approach is to boost the target domain's segmentation accuracy. Therefore, spatial reconstruction is applied exclusively to the source domain images within SJT-Net, with no preprocessing performed on the target domain. After the spatial reconstruction shown in Fig. 4(a) or (b), the source, along with target domain images, exhibits consistent sizes and closely matched spatial resolutions. This suggests that every reconstructed image in the source domain, with dimensions of 512 × 512 pixels, contains category information that closely matches the target domain.

C. Global–Local Interactive Decoder

The objects in RSI typically exhibit complex structures, textures, and shapes. To tackle segmentation challenges such as confusions with small target categories, ambiguous semantic boundaries, and significant variations in object scales within remote sensing scenes, we present a global–local interactive decoder, as depicted in Fig. 5(a). Its goal is to decode the feature maps derived from adversarial training and self-training processes, thereby producing prediction maps with optimal segmentation performance. At the beginning of this decoder, the feature maps are partitioned into global and

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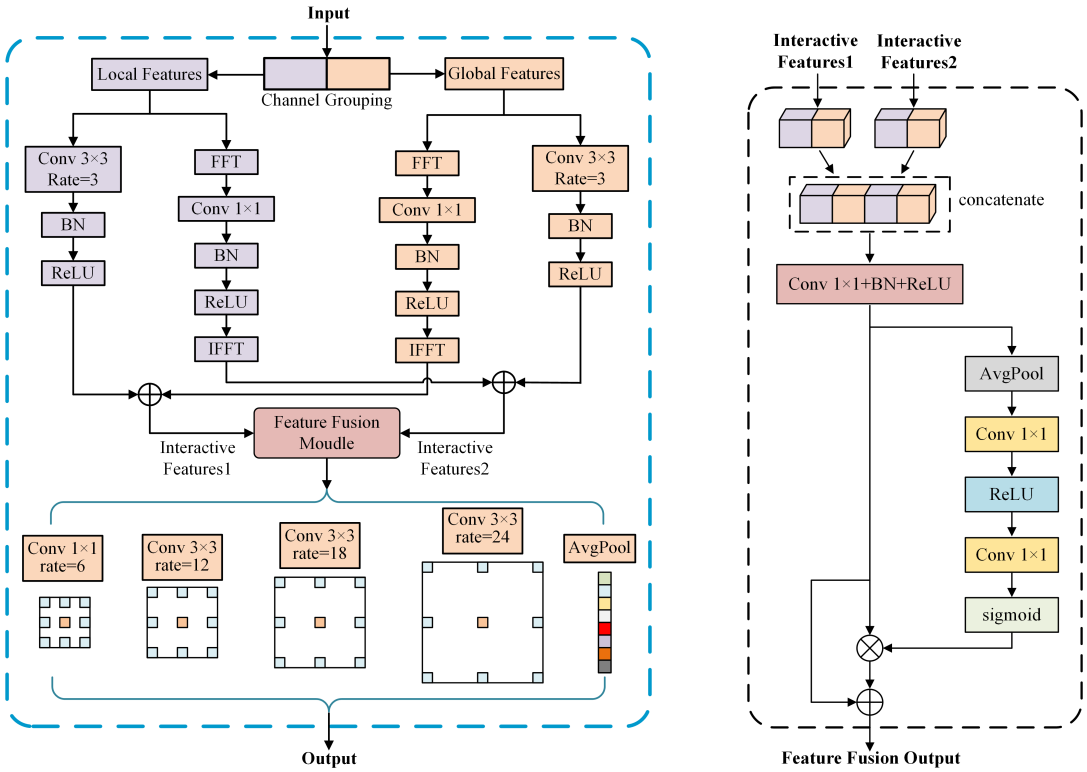


Fig. 5. (a) Overall structure of the global–local interactive decoder and (b) feature fusion module within it. The FFT in the decoder signifies the FFT, and the IFFT signifies the inverse fast Fourier transform.

local features. Initially, the input features $\mathbb{R}^{64\times64\times2048}$ are grouped along the channel dimension. Subsequently, global pooling operations and local feature processing are independently applied to each subset. Additionally, the global subset $X_{\text{global}} \in \mathbb{R}^{64\times64\times1024}$ is merged to form a comprehensive feature map, followed by the same procedure for the local subset $X_{\text{local}} \in \mathbb{R}^{64\times64\times1024}$, thereby yielding global and local feature maps $X = \{X_{\text{global}}, X_{\text{local}}\}$ across varying scales. Chi et al. [49] demonstrated that the fast Fourier transform (FFT) can expand the receptive field of conventional convolution and facilitate cross-scale feature fusion across different branches. To acquire both global and local information, we design two convolutional paths, each with a distinct receptive field. One path employs a dilation rate of 3 for atrous convolution, while the other utilizes FFT convolution. In the FFT convolution path, the feature map is first transformed from the time domain to the frequency domain using the FFT, enabling the decoder to simultaneously process both high-frequency and low-frequency feature information at a global range. This process allows the convolution to overcome the constraints of the local receptive field, effectively integrating both global and local information through frequency domain processing. After the FFT, the real part feature R_{Re} and the imaginary part feature R_{Im} are obtained, each with dimensions of $\mathbb{R}^{64\times32\times1024}$. Subsequently, point-wise multiplication, normalization, and ReLU activation are applied to the real and imaginary components. Finally, the real and imaginary components are reintegrated into time-domain features through the inverse FFT (IFFT), allowing this branch to effectively preserve global context information while simultaneously capturing local details.

Each path captures information from diverse receptive fields, enhancing the representation of a broader semantic relationship. By cross-aggregating features from both the atrous convolution path and the frequency domain transformation path, the decoder gains enhanced multiscale interaction capabilities. Furthermore, we utilize a feature fusion module [50], as depicted in Fig. 5(b), that effectively integrates feature maps from various branches, resulting in the fused features $\mathbb{R}^{64\times64\times2048}$. Introducing the attention module facilitates the automatic learning of correlations among features across various scales, guiding the model's focus toward regions rich in semantic information. By thoroughly leveraging complementary information from multiple paths, this process generates more comprehensive and accurate segmentation masks in subsequent stages. Finally, we introduce the atrous spatial pyramid pooling (ASPP) module [19], [20], [21] for feature decoding and enhance its dilation rates to [6, 12, 18, 24]. The purpose is to employ a wider spatial pyramid structure to enhance the preservation of semantic information and prevent the loss of multiscale details during feature decoding, thereby enhancing the accuracy of semantic segmentation and ultimately yielding a precise segmentation mask $\mathbb{R}^{64\times64\times K}$, where the channel K represents the number of categories.

D. Loss Function

The SJT-Net is optimized via adversarial training, including the segmentation loss $L_{\text{seg}}(X_S)$ of the source domain and the adversarial loss $L_{\text{adv}}(X_S, X_T)$. Additionally, self-training is employed, focusing on the segmentation loss $L_{\text{self}}(X_T)$ of the

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target domain. Using the cross-entropy, the segmentation loss $L_{\text{seg}}(X_S)$ of the source domain is as follows:

$$L_{\text{seg}}(X_S) = -\sum_{n=1}^{N_S}\sum_{k=1}^{K} Y_S^{(n,k)} \log \mathbf{F}\left(X_S^{(n,k)}\right) \tag{3}$$

where, in the source domain, $X_S^{(n,k)}$ signifies the kth category of the nth image, while $Y_S^{(n,k)}$ signifies the kth category of the nth label. There exist a total of N_S images and K categories. $\mathbf{F}(\cdot)$ signifies the mapping function of the segmentation backbone architecture.

Based on minimax optimization strategy, the feature extractor's adversarial loss is as follows:

$$L_{\text{adv_G}} = \min_{\mathbf{G}} \max_{\mathbf{D}} \left\{-E_{X_T \sim P(X_T)}\left[\log \mathbf{D}\left(\mathbf{G}\left(X_T\right)\right)\right]\right\}. \tag{4}$$

The discriminator's adversarial loss is as follows:

$$\begin{aligned} L_{\text{adv_D}} = \min_{\mathbf{G}} \max_{\mathbf{D}} \{ & E_{X_S \sim P(X_S)}\left[\log \mathbf{D}\left(X_S\right)\right] \\ & + E_{X_T \sim P(X_T)}\left[\log\left(1 - \mathbf{D}\left(\mathbf{G}\left(X_T\right)\right)\right)\right]\}. \end{aligned} \tag{5}$$

The feature extractor and discriminator collectively make up the adversarial loss $L_{\text{adv}}(X_S, X_T)$, as follows:

$$L_{\text{adv}}(X_S, X_T) = \lambda_1 L_{\text{adv_G}} + \lambda_2 L_{\text{adv_D}} \tag{6}$$

where λ_1 and λ_2 are set to 0.5 and 0.001, respectively. $\mathbf{G}(\cdot)$ represents the generator, and $\mathbf{D}(\cdot)$ represents the discriminator. X_S signifies the source domain's images, while X_T refers to the target one.

The first stage's adversarial training L_{stage1} is as follows:

$$L_{\text{stage1}} = L_{\text{seg}}(X_S) + L_{\text{adv}}(X_S, X_T). \tag{7}$$

The cross-entropy is used as the target domain's self-training loss $L_{\text{self}}(X_T)$ for the second stage, as follows:

$$L_{\text{stage2}} = L_{\text{self}}(X_T) = -\sum_{n=1}^{N_T}\sum_{k=1}^{K} Y_T'^{(n,k)} \log \mathbf{F}\left(X_T^{(n,k)}\right) \tag{8}$$

where, in the target domain, $X_T^{(n,k)}$ signifies the kth category of the nth image, while $Y_T'^{(n,k)}$ signifies the kth category of the nth pseudo-label. There exists a total of N_T images and K categories. $\mathbf{F}(\cdot)$ signifies the mapping function of the segmentation backbone architecture.

To outline the core optimization steps of the proposed SJT-Net model, we present a concise pseudocode that illustrates the entire training procedure, as shown in Algorithm 1.

Algorithm 1 Training Process of SJT-Net

Inputs: Source domain: Images $X_S \in \mathbb{R}^{H_S \times W_S \times 3}$, Labels $Y_S \in \mathbb{R}^{H_S \times W_S \times K}$;Target domain: Images $X_T \in \mathbb{R}^{H_T \times W_T \times 3}$ (unlabeled)

Outputs: Domain adaptation model and prediction results $\hat{Y}_T \in \mathbb{R}^{H \times W \times K}$ for target domain

1: Cross-Domain Spatial Reconstruction: Resize X_S and X_T to a shared resolution (H, W) to align semantic content. $X_S, X_T \in \mathbb{R}^{H \times W \times 3}$

2: Source Domain Pretraining:
segformer feature extractor (**G**): $X_S \rightarrow \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C}$
Global-Local Interactive Decoder as segmentation head (**H**): $\mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C} \rightarrow \mathbb{R}^{H \times W \times K_{cls}}$
Optimize **G**, **H** via cross-entropy loss: L_{seg}

3: Adversarial Training for Domain Alignment:
segformer feature extractor (**G**): $X_S \rightarrow \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C}$, $X_T \rightarrow \mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C}$
Update Discriminator (**D**): $L_{\text{adv_D}}$
Update segformer feature extractor (**G**): $L_{\text{adv_G}}$
Joint Optimization: $L_{adv} = \lambda_1 L_{\text{adv_G}} + \lambda_2 L_{\text{adv_D}}$ (λ_1, λ_2 is weight)

4: Self-Training on Target Domain:
Generate pseudo-labels $\hat{Y}_T$ for X_T using adversarial-trained model: $\mathbb{R}^{\frac{H}{8} \times \frac{W}{8} \times C} \rightarrow \mathbb{R}^{H \times W \times K}$
Optimize **G**, **H** via cross-entropy loss: L_{self}
Repeat refine $\hat{Y}_T$ iteratively

IV. Experiments

In this section, our experimental details, encompassing the remote sensing dataset descriptions, model evaluation metrics, and training parameter configurations, are described. Subsequently, by comparing with existing UDA semantic segmentation methods, we demonstrate the effectiveness of SJT-Net. Ultimately, we undertake ablation analysis to further explore the contribution of every component within SJT-Net to the improvement of model segmentation accuracy.

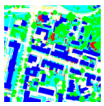
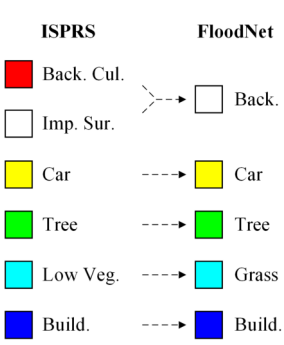
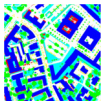
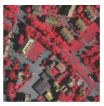
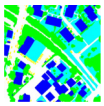
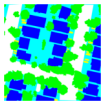
A. Dataset Details

To assess the validity of SJT-Net, this research initially conducted experiments using the RSI from the International Society for Photogrammetry and Remote Sensing (ISPRS) WG II/4: Potsdam 2D and Vaihingen 2D datasets [51]. Semantic labels are as follows: Background Clutter (Back. Cul.), Impervious Surface (Imp. Sur.), Car, Tree, Low Vegetation (Low. Veg.), and Building (Build.). As previously mentioned, RSIs originate from various regions, and the variations in GSD during data collection across different research teams present challenges for cross-domain semantic segmentation. To assess the generalization capability of SJT-Net on RSI collected by other teams, we further conduct experiments on the FloodNet dataset [52]. The semantic categories in the FloodNet dataset are redefined to align with those of the ISPRS dataset, retaining the primary categories and classifying unrelated ones as background. The redefined categories are as follows: Background (Back.), Car, Tree, Grass, and Building (Build.).

ISPRS Potsdam 2D encompasses 38 orthoimages, each with a fixed 6000 × 6000 pixels and a spatial resolution of 5 cm. As depicted in Table I, the Potsdam dataset contains two imaging modality data, with spectral bands of three-channel IRRG and RGB. When Potsdam_IRRG (or Potsdam_RGB) serves as the source domain, it is cropped to 1024 × 1024 pixel size with a 512-pixel cropping stride, separated into about 2000 training images and 1000 test images. Conversely, when Potsdam_IRRG (or Potsdam_RGB) is utilized as the target domain, it is cropped to 512 × 512 pixel size with a 512-pixel

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TABLE I
DETAILS OF THE CROSS-DOMAIN DATASET IN THE EXPERIMENT

Dataset	Size	Spatial Resolution	Band	Amount	Image	Label	Category
Potsdam	6000 × 6000	5 cm	IRRG	38			
			RGB	38			
Vaihingen	2000 × 2500	9 cm	IRRG	33			
FloodNet	4600 × 3000	3.75 cm	RGB	284			

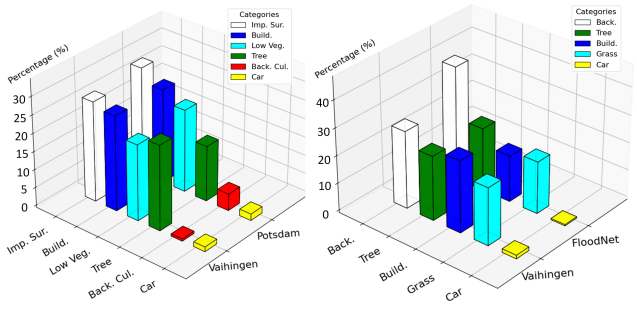


Fig. 6. Category distribution of cross-domain datasets.

cropping stride, separated into about 3000 training images and 1300 test images.

ISPRS Vaihingen 2D encompasses 33 orthoimages, each with an average resolution of about 2000 × 2500 pixels and a spatial resolution of 9 cm. As depicted in Table I, the Vaihingen dataset contains spectral bands of a three-channel IRRG. Vaihingen_IRRG is cropped to a 512 × 512 pixel size with a 256-pixel cropping stride, which functions as the source or target domain, separated into about 1200 training images and 500 test images.

FloodNet encompasses 284 images, each with an average resolution of about 4600 × 3000 pixels and a spatial resolution of 3.75 cm. As depicted in Table I, the FloodNet dataset contains spectral bands of three-channel RGB. FloodNet_RGB is cropped to a 512 × 512 pixel size with a 512-pixel cropping stride, separated into about 7700 training images and 1700 test images.

Fig. 6 illustrates the pixel distribution across each category for different datasets. These datasets exhibit a common issue of class imbalance, with significantly lower proportions in the "Back. Clu." and "Car" categories compared to others. The proportion of pixels for the "Back. Clu." category in the Vaihingen dataset and the "Car" category in the FloodNet dataset is virtually insignificant. This imbalance stems from factors like geographical location and terrain of aerial photography areas, posing some challenges in the UDA segmentation task across remote sensing scenes.

B. Model Evaluation Metrics

This research evaluates the effectiveness of semantic segmentation for cross-domain remote sensing scenes by using three different metrics, namely mIoU, m$F1$, and overall accuracy (OA). Similar evaluation metrics are applied during experimental assessments within studies [37], [38].

For each class k among the six different categories in the cross-domain dataset of this paper, the corresponding IoU and $F1$ scores are calculated as follows:

$$\mathrm{IoU}_k = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{9}$$

$$F1_k = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{10}$$

mIoU involves computing the means of IoU for all categories, and m$F1$ involves computing the means of $F1$ for all categories. The corresponding formulas are as follows:

$$\mathrm{mIoU} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{IoU}_k \tag{11}$$

$$\mathrm{m}F1 = \frac{1}{K}\sum_{k=1}^{K} F1_k. \tag{12}$$

OA measures the percentage of accurately classified pixels relative to the overall pixel count, as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{13}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative pixels, respectively.

C. Training Parameter Configurations

All experiments in this research were conducted using PyTorch version 1.13.1. The experimental device is an

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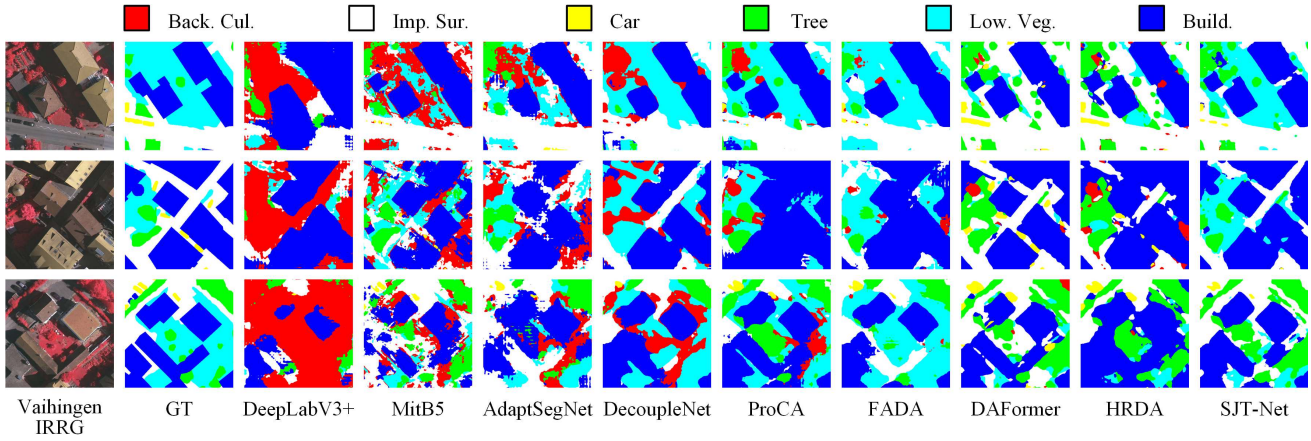


Fig. 7. Segmentation maps of varied models on the Pots_RGB → Vai_IRRG task.

NVIDIA GeForce RTX 8000 with 48 GB of VRAM and an i9 12900K processor. With an initial learning rate of $5e^{-4}$, momentum of 0.9, and weight decay of $1e^{-4}$, our model employed the stochastic gradient descent (SGD) optimizer for training the segmentation backbone architecture. The Adam optimizer was used to train the discriminator, setting an initial learning rate of $5e^{-4}$, and momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The training batch size was 4, while the testing batch size was 1.

D. Comparative Experiments

To show the excellence of the SJT-Net, we conducted four cross-domain semantic segmentation experiments, as follows.

① Source Domain: Potsdam_RGB, Target Domain: Vaihingen_IRRG, denoted as: Pots_RGB → Vai_IRRG

② Source Domain: Potsdam_IRRG, Target Domain: Vaihingen_IRRG, denoted as: Pots_IRRG → Vai_IRRG

③ Source Domain: Vaihingen_IRRG, Target Domain: Potsdam_RGB, denoted as: Vai_IRRG → Pots_RGB

④ Source Domain: Vaihingen_IRRG, Target Domain: FloodNet_RGB, denoted as: Vai_IRRG → Flood_RGB

The cross-domain experiment excludes Vai_IRRG → Pots_IRRG. This is because experiments ① and ② demonstrate superior result using similar bands, consistent with findings by Ma et al. [38] and Li et al. [1].

In previous UDA research, the DeepLab has commonly been used as the baseline model, while with less usage of the Transformer. Thus, this research contrasts two segmentation networks: DeepLabv3+ [21] and Segformer [27]. In addition, this research compares the existing UDA methods, including AdaptSegNet [13], FADA [14], DecoupleNet [15], ProCA [16], DAFormer [17], and HRDA [18]. By evaluating the output metrics of various approaches in cross-domain tasks, the proposed SJT-Net is demonstrated to be superior.

1) Cross-Domain Segmentation Comparative Analysis of Pots_RGB → Vai_IRRG: Table II shows the comparative experimental evaluation outcomes of Pots_RGB → Vai_IRRG, while Fig. 7 exhibits the visual segmentation outcomes of it. In Table II, A signifies the baseline as DeepLabv3+, B signifies the baseline as Segformer, C signifies the domain strategy as adversarial training, and D signifies the domain strategy as self-training. Table II indicates that the proposed SJT-Net surpasses the suboptimal model in overall evaluation indicators compared to the current SOTA, specifically manifested as mean IoU (mIoU) exceeding 4.53%, mean $F1$ (m$F1$) exceeding 3.15%, and OA exceeding 5.52%, respectively. While some categories did not achieve the highest IoU, these categories still attained high scores. The SJT-Net model performed best in the imbalanced category "Back. Cul." and the common category "Low. Veg." The SJT-Net model underperformed on the categories "Imp. Sur.," "Car," "Tree," and "Build.," primarily due to pseudo-label generation within an adversarial training framework. The model lacks a mechanism to handle the boundaries between semantic categories, resulting in misclassifications, such as "Imp. Sur.," "Car." and "Build." being mistaken for "Back. Cul.," and "Tree" for "Low. Veg." These categories share high visual similarity, making them difficult to distinguish without clear boundaries. As subsequent self-training iterations rely on these erroneous pseudo-labels, misclassification is compounded. Consequently, the SJT-Net model shows improved performance in "Back. Cul." and "Low. Veg." relative to other models, but its performance in other categories remains suboptimal. Despite this, the SJT-Net model enhances recognition for imbalanced categories, achieving relatively balanced overall performance without severe errors in any category. The segmentation visualization depicted in Fig. 7 further illustrates that the SJT-Net model closely matches the ground truth (GT) labels. In the absence of UDA strategies, comparing the two baseline models, Segformer outperforms DeepLabv3+ in terms of IoU in a single category, as well as overall mIoU, m$F1$, and OA. Overall, considering Table II and Fig. 7, the Transformer architecture's strong segmentation performance is far superior to DeepLabv3+, and cross-domain segmentation experiments using domain adaptation strategies generally outperform those without domain adaptation strategies.

Although AdaptSegNet employs a domain adaptation strategy, its effectiveness is hampered by its dependency on the antiquated FCN (DeepLabv3+), which compromises its advantages in domain alignment. Therefore, the segmentation outcomes of this method are marginally inferior to those of the experiment in which the baseline is Segformer without utiliz-

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TABLE II
EVALUATION OUTCOMES OF VARIED APPROACHES ON THE POTS_RGB → VAI_IRRG TASK

Methods	Baseline		UDA Strategy		Class IoU %						mIoU %	mF1 %	OA %
	A	B	C	D	Back. Cul.	Imp. Sur.	Car	Tree	Low. Veg.	Build.			
DeepLabv3+	✓	×	×	×	0.62	33.49	29.93	37.56	1.99	54.73	26.39	34.22	52.59
Segformer	×	✓	×	×	1.15	45.08	41.16	55.11	17.46	61.19	36.86	46.84	57.02
AdaptSegNet	✓	×	✓	×	1.27	46.63	30.01	53.79	16.51	58.26	34.41	43.86	56.26
DecoupleNet	✓	×	✓	×	1.51	49.67	32.76	44.77	30.17	72.36	38.54	47.72	58.38
ProCA	✓	×	✓	×	10.44	47.39	47.45	60.12	21.25	71.98	43.11	52.99	60.77
FADA	✓	×	✓	×	11.08	49.36	46.46	51.19	30.22	75.14	43.91	54.06	61.73
DAFormer	×	✓	×	✓	16.37	57.30	41.75	**69.00**	15.93	**84.81**	47.53	60.14	63.62
HRDA	×	✓	×	✓	16.27	**68.32**	**50.31**	66.94	26.70	73.26	50.30	62.47	65.73
SJT-Net	×	✓	✓	✓	**50.42**	59.20	43.68	61.98	**32.84**	80.85	**54.83**	**65.62**	**72.25**

TABLE III
EVALUATION OUTCOMES OF VARIED APPROACHES ON THE POTS_IRRG → VAI_IRRG TASK

Methods	Baseline		UDA Strategy		Class IoU %						mIoU %	mF1 %	OA %
	A	B	C	D	Back. Cul.	Imp. Sur.	Car	Tree	Low. Veg.	Build.			
DeepLabv3+	✓	×	×	×	1.09	33.01	34.68	56.45	16.89	58.62	33.46	42.90	55.43
Segformer	×	✓	×	×	2.41	49.61	42.88	59.42	31.86	66.44	42.10	52.60	59.53
AdaptSegNet	✓	×	✓	×	1.47	47.47	28.49	51.07	27.81	58.34	35.78	45.64	56.70
DecoupleNet	✓	×	✓	×	7.15	48.40	43.37	41.06	28.97	73.80	40.46	50.39	59.06
ProCA	✓	×	✓	×	24.84	55.15	40.19	59.68	35.41	74.77	48.34	60.91	64.42
FADA	✓	×	✓	×	23.41	59.14	47.82	60.17	43.02	76.18	51.62	63.45	66.98
DAFormer	×	✓	×	✓	23.48	**74.60**	48.64	**69.93**	51.18	82.27	58.35	71.00	76.18
HRDA	×	✓	×	✓	11.87	73.49	51.05	69.21	50.93	**83.19**	56.62	68.94	72.98
SJT-Net	×	✓	✓	✓	**54.66**	69.13	**52.03**	63.16	**51.85**	80.63	**61.91**	**74.16**	**78.81**

ing UDA. The SJT-Net model is built upon the Transformer structure, enabling the model to establish complex contextual dependencies on remote sensing categories during the extraction of features. Additionally, our model employs adversarial domain adaptation strategies to produce high-caliber pseudo-labels, laying the foundation that supports further self-training, which outperforms methods that rely solely on self-training domain adaptation strategies. Finally, we iteratively update the pseudo labels and models using self-training to boost the segmentation accuracy of the target domain. The SJT-Net model not only demonstrates robustness in segmentation networks but also comprehensively integrates the benefits of both domain adaptation strategies, complementing each other and fully leveraging adversarial training and self-training advantages, thereby yielding superior cross-domain segmentation results.

2) Cross-Domain Segmentation Comparative Analysis of Pots_IRRG → Vai_IRRG: Table III shows the comparative experimental evaluation outcomes of Pots_IRRG → Vai_IRRG, while Fig. 8 exhibits the visual segmentation outcomes of it. In Table III, A signifies the baseline as DeepLabv3+, B signifies the baseline as Segformer, C signifies the domain strategy as adversarial training, and D signifies the domain strategy as self-training. Table III indicates that the SJT-Net model surpasses the suboptimal model in overall evaluation indicators compared to the current SOTA, specifically manifested as mean IoU (mIoU) exceeding 3.56%, mean $F1$ (m$F1$) exceeding 3.16%, and OA exceeding 5.03%, respectively. While some categories did not achieve the highest IoU, these categories still attained high scores, and the SJT-Net model achieved the best results in two imbalanced categories "Back. Cul." and "Car" as well as the common category "Low. Veg." The segmentation visualization depicted in Fig. 8 also reveals that the SJT-Net model closely matches the GT labels. Similar conclusions regarding cross-domain segmentation performance can be drawn by analyzing Table III and Fig. 8, consistent with findings from previous comparative experiments.

3) Cross-Domain Segmentation Comparative Analysis of Vai_IRRG → Pots_RGB: Table IV shows the comparative experimental evaluation outcomes of Vai_IRRG → Pots_RGB, while Fig. 9 exhibits the visual segmentation outcomes of it. In Table IV, A signifies the baseline as DeepLabv3+, B signifies the baseline as Segformer, C signifies the domain strategy as adversarial training, and D signifies

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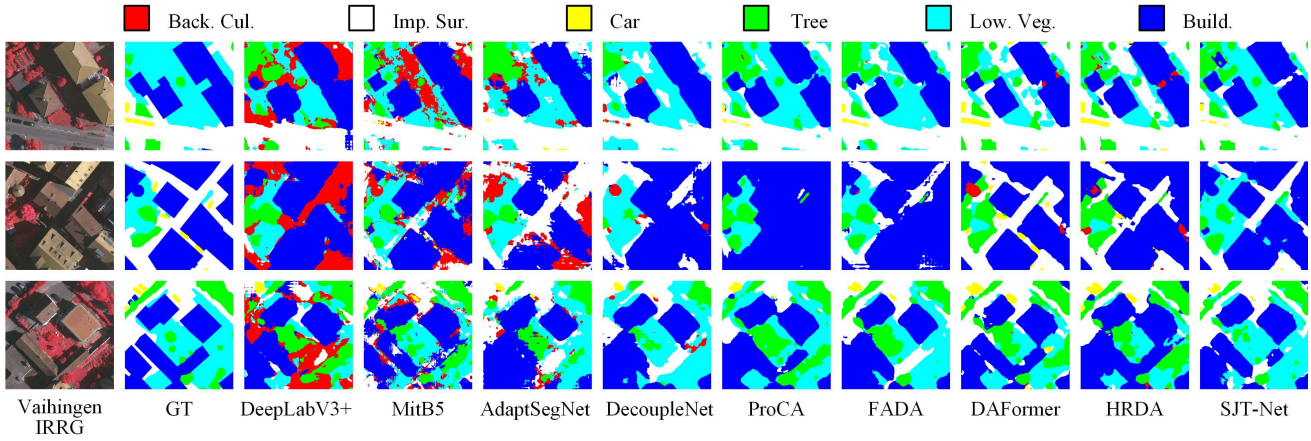


Fig. 8. Segmentation maps of varied models on the Pots_IRRG → Vai_IRRG task.

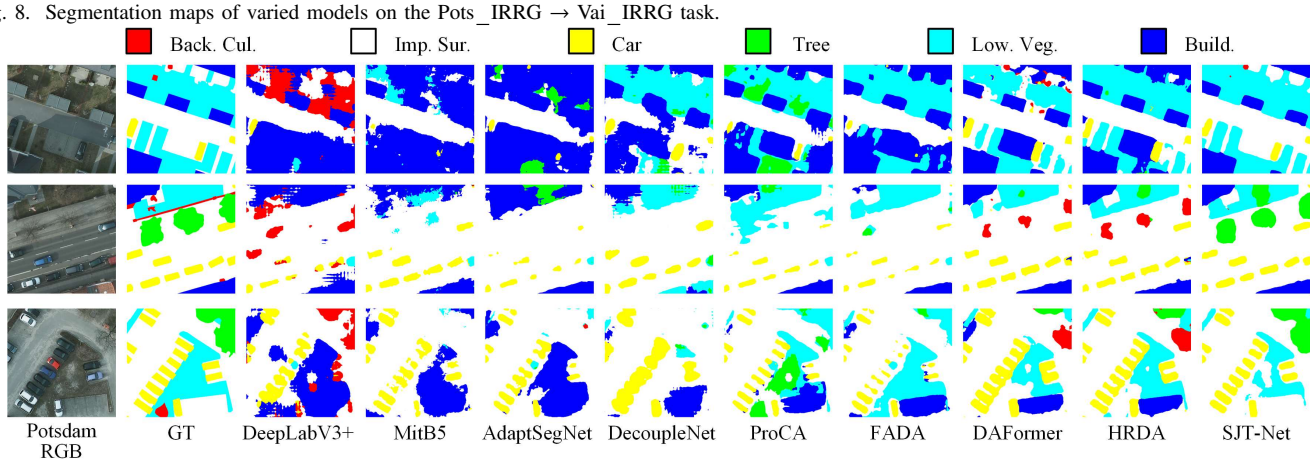


Fig. 9. Segmentation maps of varied models on the Vai_IRRG → Pots_RGB task.

TABLE IV

EVALUATION OUTCOMES OF VARIED APPROACHES ON THE VAI_IRRG → POTS_RGB

Methods	Baseline	UDA Strategy			Class IoU %						mIoU %	mF1 %	OA %
	A	B	C	D	Back. Cul.	Imp. Sur.	Car	Tree	Low. Veg.	Build.			
DeepLabv3+	✓	×	×	×	4.07	34.29	41.20	13.82	5.84	29.45	21.45	26.67	48.34
Segformer	×	✓	×	×	15.44	36.90	61.48	22.81	18.98	33.72	31.56	38.32	53.19
AdaptSegNet	✓	×	✓	×	16.50	44.35	58.68	29.69	9.25	42.41	33.48	43.12	56.08
DecoupleNet	✓	×	✓	×	20.22	47.53	67.26	24.23	29.83	43.63	38.78	47.83	59.04
ProCA	✓	×	✓	×	25.50	44.19	62.56	21.59	36.47	60.40	41.79	51.36	59.17
FADA	✓	×	✓	×	26.68	48.91	55.08	26.22	38.60	57.81	42.22	52.66	60.05
DAFormer	×	✓	×	✓	20.47	65.68	64.76	51.91	**58.34**	76.65	56.30	68.12	72.92
HRDA	×	✓	×	✓	16.13	70.63	65.35	53.07	57.84	**77.02**	56.67	69.88	73.27
SJT-Net	×	✓	✓	✓	**29.53**	**70.65**	**71.67**	**61.41**	56.79	75.58	**60.94**	**73.37**	**77.93**

the domain strategy as self-training. Table IV indicates that the SJT-Net model surpasses the suboptimal model in overall evaluation indicators compared to the current SOTA, specifically manifested as mean IoU (mIoU) exceeding 4.27%, mean $F1$ (m$F1$) exceeding 3.49%, and OA exceeding 4.66%, respectively. While some categories did not achieve the highest IoU, these categories still attained high scores, and the SJT-Net model achieved the best results in two imbalanced categories "Back. Cul." and "Car" as well as two common categories "Imp. Sur." and "Tree." The segmentation visualization depicted in Fig. 9 also reveals that the SJT-Net model closely matches the GT labels. Similar conclusions regarding cross-domain segmentation performance can be drawn by analyzing Table IV and Fig. 9, consistent with findings from the previous two comparative experiments.

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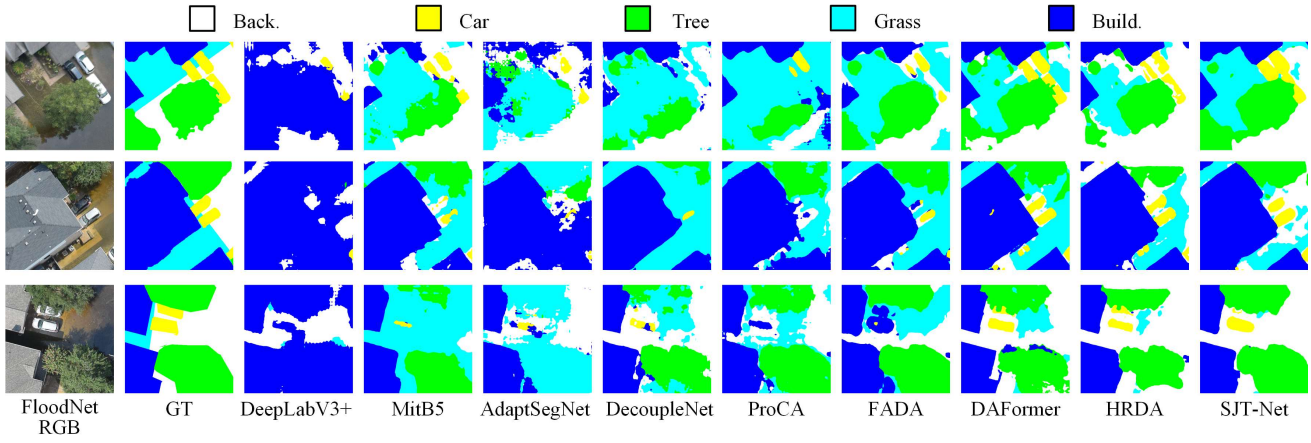


Fig. 10. Segmentation maps of varied models on the Vai_IRRG → Flood_RGB task.

TABLE V
EVALUATION OUTCOMES OF VARIED APPROACHES ON THE VAI_IRRG → FLOOD_RGB

Methods	Baseline		UDA Strategy		Class IoU %					mIoU %	mF1 %	OA %
	A	B	C	D	Back.	Car	Tree	Grass	Build.			
DeepLabv3+	✓	×	×	×	18.11	37.68	23.56	18.82	45.60	28.75	31.68	36.65
Segformer	×	✓	×	×	20.42	42.15	35.02	26.19	57.40	36.24	42.20	52.97
AdaptSegNet	✓	×	✓	×	20.62	43.32	22.60	28.17	57.54	34.45	38.25	46.17
DecoupleNet	✓	×	✓	×	36.91	41.96	50.05	30.49	72.59	46.40	60.37	65.44
ProCA	✓	×	✓	×	34.94	41.73	62.62	31.11	69.65	48.01	62.73	68.04
FADA	✓	×	✓	×	35.40	42.11	64.38	32.95	70.82	49.13	64.57	69.76
DAFormer	×	✓	×	✓	38.53	**44.88**	60.80	35.51	73.56	50.66	66.07	71.00
HRDA	×	✓	×	✓	44.33	38.27	63.69	34.89	77.43	51.72	67.68	71.92
SJT-Net	×	✓	✓	✓	**45.28**	44.11	**65.18**	**36.27**	**78.66**	**53.90**	**69.88**	**75.74**

4) Cross-Domain Segmentation Comparative Analysis of Vai_IRRG → Flood_RGB: Table IV shows the comparative experimental evaluation outcomes of Vai_IRRG → Flood_RGB, while Fig. 10 exhibits the visual segmentation outcomes of it. In Table V, A signifies the baseline as DeepLabv3+, B signifies the baseline as Segformer, C signifies the domain strategy as adversarial training, and D signifies the domain strategy as self-training. Table V indicates that the SJT-Net model surpasses the suboptimal model in overall evaluation indicators compared to the current SOTA, specifically manifested as mean IoU (mIoU) exceeding 2.18%, mean $F1$ (m$F1$) exceeding 2.20%, and OA exceeding 3.82%, respectively. The SJT-Net achieved the best results in "Back.," "Tree," "Grass," and "Build." While some categories did not achieve the highest IoU, these categories still attained high scores. The segmentation visualization depicted in Fig. 10 also reveals that the SJT-Net model closely matches the GT labels. Similar conclusions regarding cross-domain segmentation performance can be drawn by analyzing Table V and Fig. 10, consistent with findings from the previous three comparative experiments.

5) Visual Analysis of Confusion Matrices in Four Cross-Domain Segmentation Comparative Experiments: This section presents the visualization of confusion matrices for four sets of cross-domain RSI semantic segmentation comparative experiments, depicting both SJT-Net and suboptimal models, as illustrated in Fig. 11. Specifically, Fig. 11(a1) and (a2) shows the confusion matrices for SJT-Net and the suboptimal model, respectively, in the Pots_RGB → Vai_IRRG task. Similarly, Fig. 11(b1) and (b2) illustrates the confusion matrices for the same comparative models in the Pots_IRRG → Vai_IRRG task. Likewise, Fig. 11(c1) and (c2) shows the matrices for the Vai_IRRG → Pots_RGB task, and 11(d1) and (d2) for the Vai_IRRG → Flood_RGB task. By comparing the confusion matrices between the SJT-Net model and the suboptimal model, we gain a more intuitive understanding of their performance disparities in diverse cross-domain semantic segmentation tasks. The confusion matrix analysis, depicted in Fig. 11, reveals that the SJT-Net model exhibits predominantly high values along its diagonal elements, contrasting with lower values in its off-diagonal counterparts. The confusion matrix of SJT-Net exhibits a higher concentration of its elements near the main diagonal. Conversely, the elements of suboptimal models display a more scattered distribution, a trend consistently observed across all four cross-domain segmentation experiments. This also demonstrates that the SJT-Net model possesses the most comprehensive segmentation performance and superior generalization perfor-

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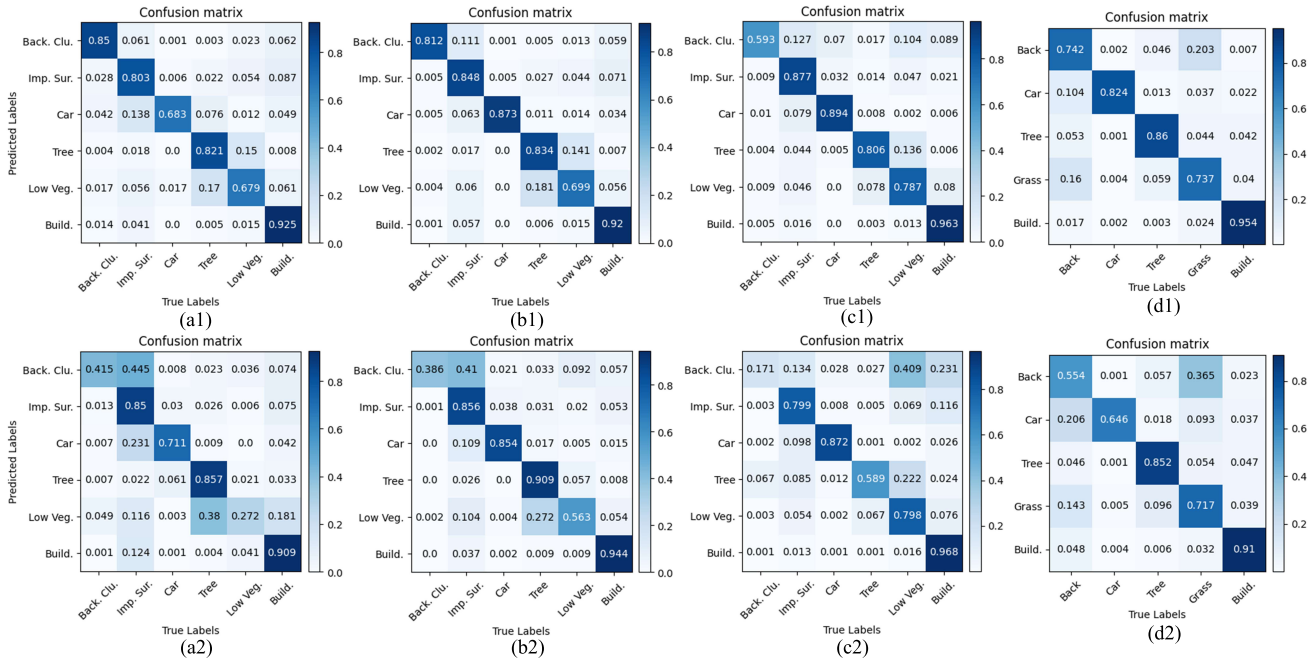


Fig. 11. Confusion matricesbetween SJT-Net and suboptimal models in four cross-domain segmentation comparative experiments.

TABLE VI

ABLATION OUTCOMES OF VARIED IMPROVED MODULES ON FOUR CROSS-DOMAIN SEGMENTATION TASKS

Tasks	Cross-Domain Spatial Reconstruction (CDSR)	Segformer Feature Extractor (SFE)	Global-Local Interactive Decoder (GLID)	mIoU %	mF1 %	OA %
Pots_RGB → Vai_IRRG	×	✓	✓	53.42	64.46	68.61
	✓	×	✓	46.57	56.41	62.64
	✓	✓	×	51.74	63.72	67.72
	✓	✓	✓	**54.83**	**65.62**	**72.25**
Pots_IRRG → Vai_IRRG	×	✓	✓	59.51	72.77	76.71
	✓	×	✓	52.26	63.83	68.09
	✓	✓	×	56.95	70.04	73.48
	✓	✓	✓	**61.91**	**74.16**	**78.81**
Vai_IRRG → Pots_IRRG	×	✓	✓	58.41	71.42	76.42
	✓	×	✓	50.68	62.71	65.97
	✓	✓	×	55.72	66.23	72.56
	✓	✓	✓	**60.94**	**73.37**	**77.93**
Vai_IRRG → Flood_RGB	×	✓	✓	52.22	68.18	72.19
	✓	×	✓	49.37	65.21	70.09
	✓	✓	×	50.85	66.32	71.33
	✓	✓	✓	**53.90**	**69.88**	**75.74**

mance when handling cross-domain semantic segmentation tasks.

E. Ablation Experiment

1) Ablation Experiments of Different Improved Modules: In demonstrating how each improved module in SJT-Net impacts model performance, various modules are applied in four cross-domain RSI semantic segmentation tasks for ablation analysis, as depicted in Table VI. CDSR denotes cross-domain spatial reconstruction, SFE denotes Segformer feature extractor, and GLID denotes global–local interactive decoder. In the Pots_RGB → Vai_IRRG task, compared with the SJT-Net model, when the CDSR is not used, the mIoU value drops by 1.41%. When the SFE is not used and ResNet101 is used as the feature extractor, the mIoU value drops by 8.26%. When the GLID is not used and the ASPP module is used to directly perform feature decoding, the mIoU value drops by 2.99%. Similarly, in the Pots_IRRG → Vai_IRRG task, the same ablation operations, when compared to the SJT-Net model, lead to a decrease in mIoU by 2.4%, 9.65%, and

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TABLE VII

ABLATION OUTCOMES OF VARIED DOMAIN ADAPTATION STRATEGIES ON FOUR CROSS-DOMAIN SEGMENTATION TASKS

Tasks	Adversarial Training	Self-training	mIoU %	mF1 %	OA %
Pots_RGB → Vai_IRRG	×	×	40.79	50.55	59.09
	✓	×	49.48	61.27	64.78
	×	✓	45.17	55.32	62.46
	✓	✓	**54.83**	**65.62**	**72.25**
Pots_IRRG → Vai_IRRG	×	×	47.15	58.92	62.99
	✓	×	57.86	70.76	75.07
	×	✓	54.25	64.91	71.66
	✓	✓	**61.91**	**74.16**	**78.81**
Vai_IRRG → Pots_IRRG	×	×	41.26	51.07	59.12
	✓	×	54.48	65.29	71.87
	×	✓	50.16	62.30	65.35
	✓	✓	**60.94**	**73.37**	**77.93**
Vai_IRRG → Flood_RGB	×	×	38.24	45.20	54.56
	✓	×	48.16	63.05	68.23
	×	✓	45.59	59.68	64.88
	✓	✓	**53.90**	**69.88**	**75.74**

TABLE VIII

ABLATION OUTCOMES OF VARIED SPATIAL RECONSTRUCTION STRATEGIES ON FOUR CROSS-DOMAIN SEGMENTATION TASKS

Tasks	Gaussian Pyramid	Bilinear Interpolation	mIoU %	mF1 %	OA %
Pots_RGB → Vai_IRRG	×	×	53.42	64.46	68.61
	✓	×	53.94	64.72	69.34
	×	✓	54.23	64.89	71.58
	✓	✓	**54.83**	**65.62**	**72.25**
Pots_IRRG → Vai_IRRG	×	×	59.51	72.77	76.71
	✓	×	61.20	73.47	78.23
	×	✓	61.50	73.76	78.75
	✓	✓	**61.91**	**74.16**	**78.81**
Vai_IRRG → Pots_IRRG	×	×	58.41	71.42	76.42
	✓	×	60.22	72.84	77.26
	×	✓	60.40	73.11	77.68
	✓	✓	**60.94**	**73.37**	**77.93**
Vai_IRRG → Flood_RGB	×	×	52.22	68.18	72.19
	✓	×	53.14	69.28	72.37
	×	✓	53.55	69.57	73.53
	✓	✓	**53.90**	**69.88**	**75.74**

4.96%, respectively. In the Vai_IRRG → Pots_RGB task, the same ablation operations lead to a decrease in mIoU by 2.53%, 10.26%, and 5.22%, respectively, compared to the SJT-Net model. In the Vai_IRRG → Flood_RGB task, the same ablation operations lead to a decrease in mIoU by 1.68%, 4.53%, and 3.05%, respectively, compared to the SJT-Net model. The evaluation of four cross-domain segmentation tasks shows that the improved modules enhance the SJT-Net's performance to varying degrees. Specifically, the SFE provides the most significant improvement, followed by the GLID, and then the CDSR.

2) Ablation Experiments of Different Domain Adaptation Strategies: This section presents ablation studies across four remote sensing tasks to assess the impact of various domain adaptation strategies on the cross-domain segmentation performance of SJT-Net. The backbone of the SJT-Net model consists of an improved segformer feature extractor and a global–local interactive decoder, with the different UDA strategies detailed in Table VII. The ablation analysis without domain adaptation involves training the segmentation backbone on source domain samples and directly applying the learned weights to the target domain. In the adversarial training ablation, the backbone is trained on the source domain, with the learned weights capturing the target domain's semantic characteristics. Features from both domains are aligned using a discriminator, and adversarial features are subsequently decoded into segmentation maps. In the self-training ablation, the backbone is trained on the source domain, and the learned weights are used to predict the target domain. These predictions are treated as pseudo-labels, which are iteratively refined to optimize segmentation performance on the target domain. In the Pots_RGB → Vai_IRRG task, when no domain adaptation strategy is applied, the mIoU value decreases by 14.04% compared to the SJT-Net model. With the use of adversarial training for feature-level domain alignment, the mIoU value decreases by 5.35%. When using self-training for feature-level domain alignment, the mIoU value decreases by 9.66%. Similarly, in the Pots_IRRG → Vai_IRRG task, the same ablation operations lead to a decrease in mIoU by 14.76%, 4.05%, and 7.66%, respectively. In the Vai_IRRG → Pots_RGB task, the same ablation operations lead to a decrease in mIoU by 19.68%, 6.46%, and 10.78%, respectively. In the Vai_IRRG → Flood_RGB task, the same ablation operations lead to a decrease in mIoU by 15.66%, 5.74%, and 8.31%, respectively. Cross-domain RSI often exhibits significant class imbalance and scale variations. Evaluation across four cross-domain segmentation tasks reveals that direct segmentation, without domain adaptation, impairs the model's generalization ability, leading to suboptimal performance. Although both adversarial and self-training domain adaptation techniques can improve segmentation, their individual benefits are limited and fail to fully harness their potential. To overcome this limitation, we propose SJT-Net, a model that integrates adversarial and self-training strategies. Adversarial training generates precise pseudo-labels, which serve as reliable pseudo-supervision for the self-training process. This combined approach enhances cross-domain segmentation performance by leveraging the strengths of both strategies.

3) Ablation Experiments of Different Spatial Reconstruction Strategies: This section demonstrates the effect of spatial reconstruction strategies in SJT-Net on cross-domain segmentation performance through ablation experiments conducted on four RSI tasks. Various image resampling techniques are employed to align the spatial resolution of source domain

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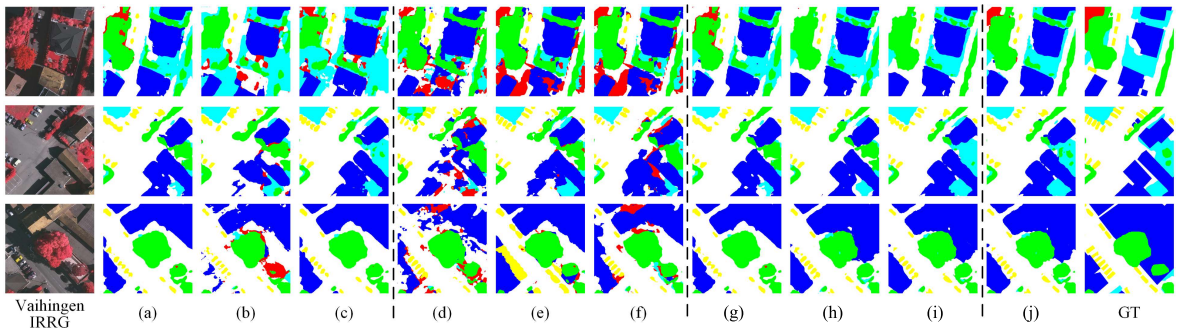


Fig. 12. Segmentation maps of ablation experiment for Pots_RGB → Vai_IRRG task.

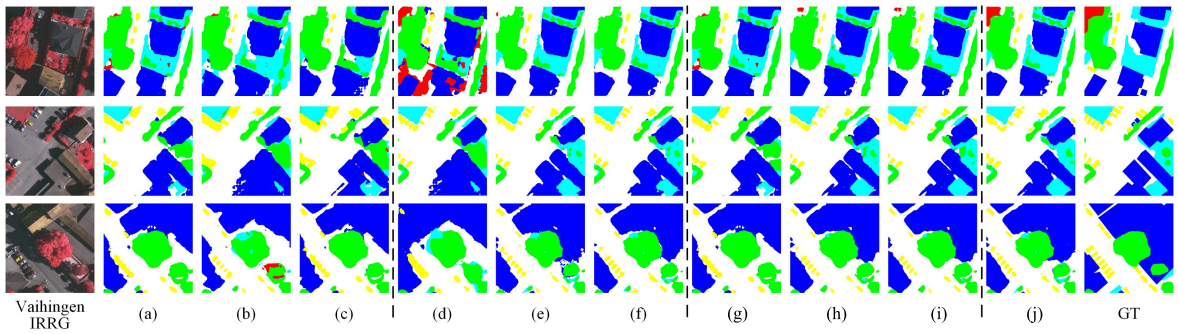


Fig. 13. Segmentation maps of ablation experiment for Pots_IRRG → Vai_IRRG task.

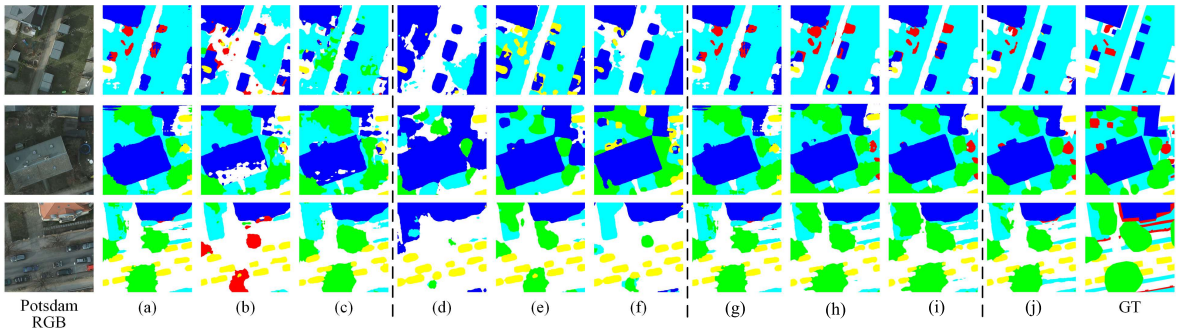


Fig. 14. Segmentation map of ablation experiment for Vai_IRRG → Pots_RGB task.

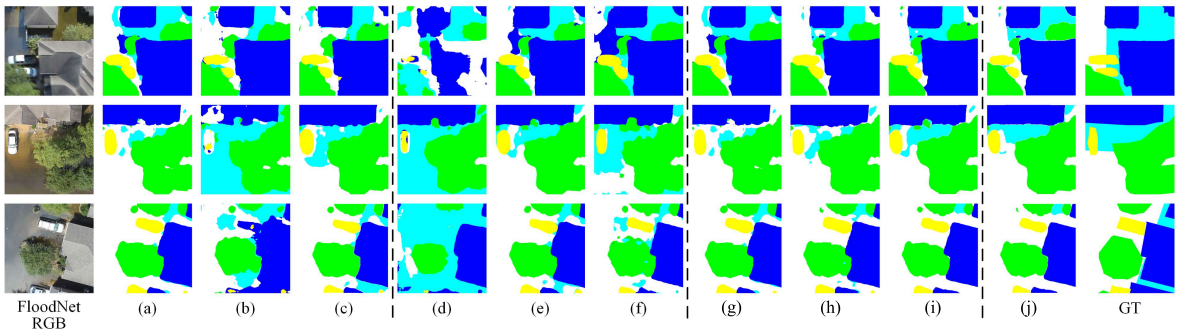


Fig. 15. Segmentation map of ablation experiment for Vai_IRRG → Flood_RGB task.

images with that of the target domain, as detailed in Table VIII. In the absence of spatial reconstruction, both domain images are cropped to 512×512 pixels, resulting in resolution discrepancies between the domains, as illustrated in Table I. In the Pots_RGB → Vai_IRRG task, without spatial reconstruction, the mIoU decreases by 1.41%. Gaussian pyramid sampling reduces this decrease to 0.89%, while bilinear interpolation results in a 0.60% decrease. Similarly, in the Pots_IRRG

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→ Vai_IRRG task, the same ablation operations lead to a decrease in mIoU by 2.40%, 0.71%, and 0.41%, respectively. In the Vai_IRRG → Pots_RGB task, the same ablation operations lead to a decrease in mIoU by 2.53%, 0.72%, and 0.54%, respectively. In the Vai_IRRG → Flood_RGB task, the same ablation operations lead to a decrease in mIoU by 1.68%, 0.76%, and 0.35%, respectively. Analysis of evaluation results from four cross-domain segmentation tasks demonstrates that the SJT-Net model enhances performance through cross-domain spatial reconstruction. Bilinear interpolation outperforms Gaussian pyramid sampling slightly, and their combination yields smoother images while preserving spatial details, thereby enhancing image reconstruction quality and further improving overall segmentation performance.

4) Visualization of Ablation Experiments: This section presents visual segmentation maps from ablation experiments conducted across four cross-domain tasks. Specifically, Figs. 12–15 depict the following tasks: Pots_RGB → Vai_IRRG (Fig. 12), Pots_IRRG → Vai_IRRG (Fig. 13), Vai_IRRG → Pots_RGB (Fig. 14), and Vai_IRRG → Flood_RGB (Fig. 15). The ablation methods are represented by legends with matching subscripts in Figs. 12–15. Legends (a)–(c) correspond to ablation experiments on improved modules (Section IV-E1), where (a) indicates no cross-domain spatial reconstruction, (b) indicates no Segformer feature extractor, and (c) indicates no global–local interactive decoder. Legends (d)–(f) refer to domain adaptation strategies (Section IV-E2), where (d) indicates no domain adaptation, (e) indicates adversarial training, and (f) denoting self-training. Legends (g) and (h) refer to spatial reconstruction strategies (Section IV-E3), with (g) utilizing Gaussian pyramid sampling and (h) using bilinear interpolation. Legend (i) indicates the proposed SJT-Net, while "GT" indicates the ground truth labels. The visual results in Figs. 12–15 show that SJT-Net closely aligns with the GT across all tasks, showing superior segmentation accuracy.

V. Conclusion

To address domain distribution imbalance in UDA RSI, this research combines spatial reconstruction and joint training with the Transformer network (SJT-Net). The proposed SJT-Net aligns spatial scales through spatial reconstruction, while synergistically unifying adversarial training and self-training, which overcomes adversarial training's limited segmentation modeling capacity and self-training's dependency on initial pseudo-label quality. Our global–local interaction decoder captures multiscale features and complex spatial dependencies. Comprehensive experiments on four cross-domain benchmarks demonstrate SJT-Net's superiority over SOTA methods, with ablation studies validating the effectiveness of critical modules. In practical remote sensing interpretation, cross-domain segmentation from a single source to multiple targets and the utilization of multimodal information are both essential. However, these factors have not been addressed within our framework. Future work will focus on developing multitarget UDA frameworks and multimodal fusion strategies to enhance RSI interpretation utility.

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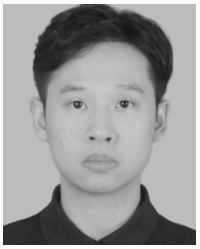
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